PHYS4055 Week 1 Tutorial Notes

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Question 1: Displacement of the annihilation operator

The displacement operator is defined by

$$\hat{D}(\alpha) = \exp\left(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\right), \tag{1}$$

where α is a complex number and

$$[\hat{a}, \hat{a}^\dagger] = 1. \tag{2}$$

Use the Baker–Campbell–Hausdorff relation given by

$$\begin{aligned} e^{\hat{X}}\hat{Y}e^{-\hat{X}} = \hat{Y} &+ [\hat{X}, \hat{Y}] \\ &+ \frac{1}{2!}[\hat{X}, [\hat{X}, \hat{Y}]] \\ &+ \frac{1}{3!}[\hat{X}, [\hat{X}, [\hat{X}, \hat{Y}]]] + \cdots . \end{aligned} \tag{3}$$

for any two operators $\hat{X}$ and $\hat{Y}$, to show that

$$\boxed{\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha}. \tag{4}$$

Question 2: Coherent states as eigenstates

A coherent state is defined by

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle, \tag{5}$$

where $|0\rangle$ is the vacuum state.

Using the displacement-operator identity

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha, \tag{6}$$

show that $|\alpha\rangle$ is an eigenstate of the annihilation operator $\hat{a}$.

Question 3: Electric field of a coherent state

Consider a single travelling optical mode with electric-field operator

$$\hat{\mathbf{E}}(z,t) = \mathbf{e}E_0\left(\hat{a}e^{i\phi} + \hat{a}^\dagger e^{-i\phi}\right), \tag{7}$$

where

$$E_0 = \sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}} \tag{8}$$

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and

$$\phi = kz - \omega t. \tag{9}$$

Here, $\mathbf{e}$ is the polarization vector, V is the quantization volume, and ω is the angular frequency of the mode. For a coherent state satisfying

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \tag{10}$$

calculate the expectation value

$$\langle\hat{\mathbf{E}}(z,t)\rangle_\alpha = \langle\alpha|\hat{\mathbf{E}}(z,t)|\alpha\rangle. \tag{11}$$

Write

$$\alpha = |\alpha|e^{i\theta} \tag{12}$$

and show that the expectation value has the form of a classical oscillating electric field.